The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 March 1995

INCH-POUND

MIL-S-19500/551B
29 December 1994
SUPERSEDING
MIL-S-19500/551A(USAF)
20 August 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, TRANSIENT VOLTAGE SUPPRESSOR
TYPES 1N6461 THROUGH 1N6468 AND 1N6461US THROUGH 1N6468US
JAN, JANTX, JANTXV, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 500-watt peak pulse, power, silicon, transient voltage suppressor diodes. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500, and two levels of product assurance for each unencapsulated device type die device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figures 1, 2, 3, 4, and 5.

1.3 Maximum ratings. Maximum ratings are as shown in columns 4, 6, and 7 of table III herein and as follows:

P_R = 2.5 W (T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750) (derate at 16.7 mW/°C for nonsurface mount devices) (10 mW/°C for surface mount devices) (see figures 3, 4, and 5).

P_{PR} = 500 W (see figure 7) at t_p = 1 ms.

I_{FSM} = 80 A(pk) at t_p = 8.33 ms (T_A = +25°C).

$-55°C \leq T_{op} \leq +175°C$; $-55°C \leq T_{STG} \leq +175°C$ (ambient).

1.4 Primary electrical characteristics. Primary electrical characteristic columns 2 and 4 of table III herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO) NASA Goddard Space Flight Center, Code 310.A Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this drawing, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions shall be as specified in MIL-S-19500.

I_P	Maximum peak pulse current at a specified condition.
$I_{(BR)}$	Reverse breakdown current at a specified condition.
T_{EC}	Endcap temperature.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1, 2, 3, 4, and 5 herein. The surface mount devices (US) shall be structurally identical to the "non US" version devices except for lead attachment.

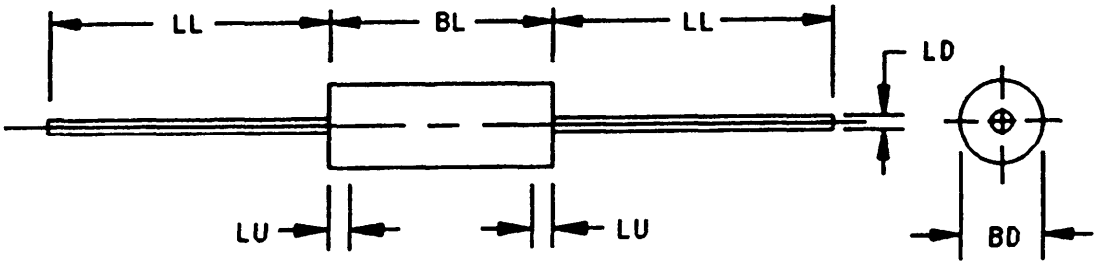
3.3.1 Construction. These devices shall be constructed utilizing metallurgically bonded-noncavity-double-plug construction between both sides of the silicon die and terminal pins. Opaque glass is not permitted.

3.3.2 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of surface mount (US) devices. For surface mount devices (US) versions only, all marking (except 3.4.2) may be omitted from the body of the device, but shall be retained on the initial container.

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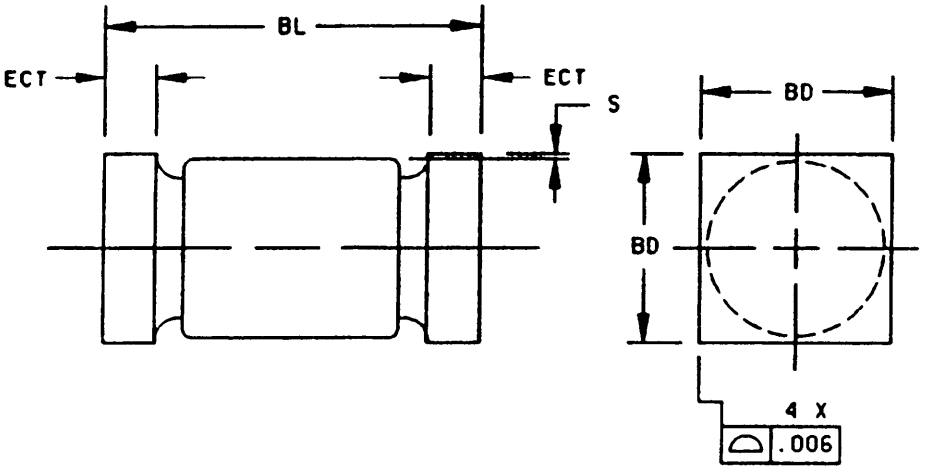


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.115	.145	2.92	3.68	
LD	.037	.042	0.94	1.07	
BL	.150	.300	3.81	7.62	3
LL	.900	1.300	22.86	33.02	
LU		.050		1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Dimension BL includes dimension LU region in which the diameter may vary from BD maximum to LD minimum.

FIGURE 1. Physical dimensions.

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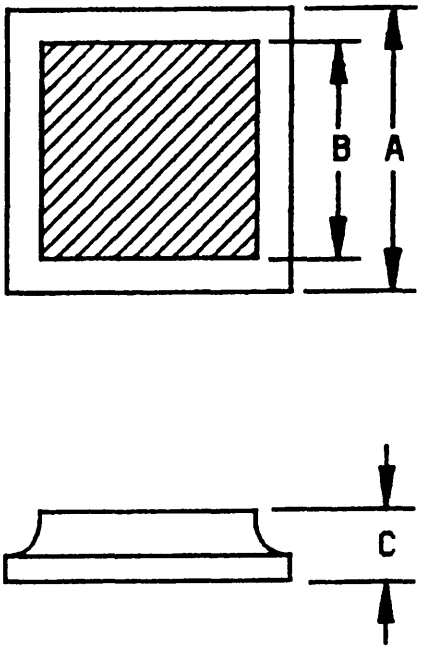


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.200	.225	5.08	5.72	
ECT	.019	.028	0.48	0.71	
S	.003		0.08		
BD	.137	.148	3.48	3.76	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions surface mount devices.

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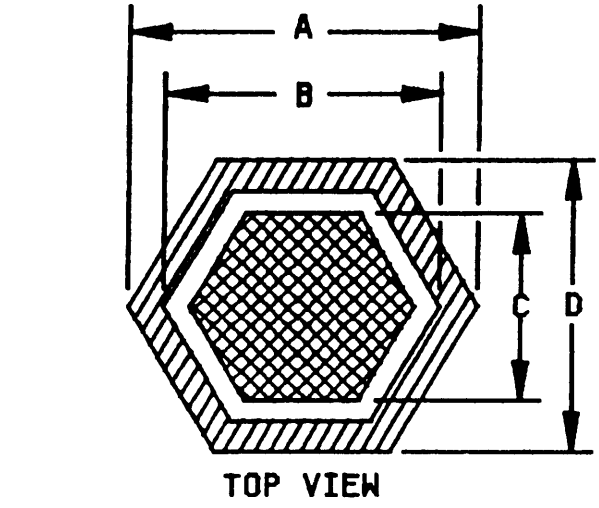
A version

Type	Inches		Millimeters	
	MIN	MAX	MIN	MAX
A	.059 sq	.065 sq	1.50 sq	1.65 sq
B	.056 sq	.060 sq	1.42 sq	1.52 sq
C	.006	.012	0.15	0.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metallization: Top (cathode) - Aluminum.
   Back (anode) - Gold.

FIGURE 3. Physical dimensions JANHC and JANKC (die).

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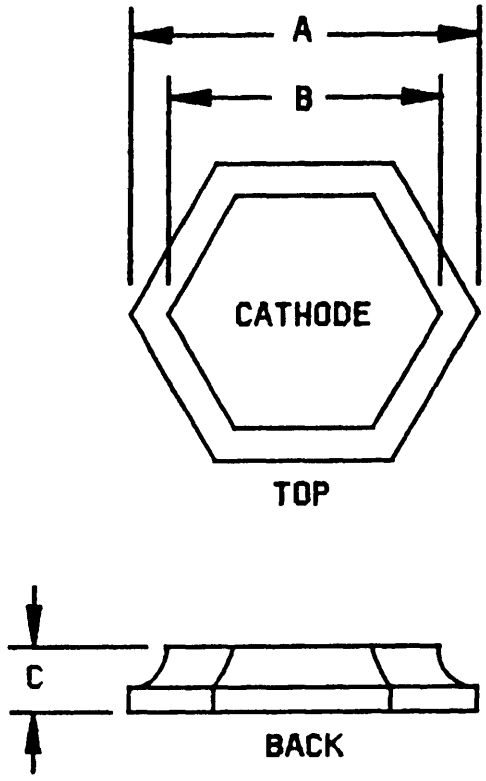
B version

Type	Inches		Millimeters	
	MIN	MAX	MIN	MAX
A	.085	.089	2.18	2.26
B	.076	.080	1.93	2.03
C	.0066	.070	1.68	1.78
D	.074	.078	1.88	1.98
E	.010	.015	0.25	0.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metallization: Top (cathode) - Aluminum.
   Back (anode) - Ti/Ni/AU.

FIGURE 4. Physical dimensions JANHC and JANKC (die).

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C version

Type	Inches		Millimeters	
	MIN	MAX	MIN	MAX
A	.085	.091	2.16	2.31
B	.072	.085	1.83	2.16
C	.007	.012	0.18	0.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metallization: Top (cathode) - Aluminum.
   Back (anode) - Gold.

FIGURE 5. Physical dimensions JANHC and JANKC (die).

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3.4.2 Marking of JANHC and JANKC die. For JANHC and JANKC die, the following marking (initial container) shall be used (example):

JANHCAM1N6461
↑↑↑↑
|||└ RHA level (see MIL-S-19500).
||└── Source of manufacturer (see figures 3, 4, and 5).
|└─── Unencapsulated.
└──── Product assurance level 1/.

1/ Two levels of product assurance levels are provided for unencapsulated devices, H and K, (see MIL-S-19500.

3.4.3 Polarity. The polarity of all device types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for surface mount (US) devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein. Lot accumulation period shall be 6 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 JANHC and JANKC die qualification. JANHC and JANKC die qualification shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JANTXV and JANTX levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
9	Not applicable
11	Not applicable
12	See 4.3.2
13	Interim electrical, delta and group A, subgroups 2 and 3 electrical parameters not applicable for this screen (performed in screen 12).

4.3.1 Screening (JANC). Screening of JANC die shall be in accordance with MIL-S-19500, appendix H.

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4.3.2 Power burn-in (HTRB) and steady-state operation life (HTRB) conditions. This test shall be conducted with the devices subjected to test conditions in the following order of events and conditions:

a. Read and record I_R and $V_{(BR)}$ at T_A = room ambient as defined in the general requirements of MIL-STD-750.

b. Pulse in accordance with 4.5.1b 20 times (screening and group B operation life test) and 100 times (group C) at T_A = room ambient as defined in the general requirements of MIL-STD-750.

c. Read and record I_R at T_A = room ambient as defined in the general requirements of MIL-STD-750. Remove defective devices, and record the number of failures.

d. Apply the working peak reverse voltage (V_{RWM}) (column 4 of table III) at an ambient temperature of +125°C as follows:

(1) 96 hours (JANTX and JANTXV) for the screening test.

(2) 340 hours (JANTX and JANTXV) for group B, steady-state operation life test.

(3) 1,000 hours for group C, steady-state operation life test.

e. Read and record I_R and $V_{(BR)}$ at T_A = room ambient as defined in the general requirements of MIL-STD-750. Devices with ΔI_R > 50 percent (100 percent for steady-state operation life) of initial value, or 20 percent of column 5 of table III, whichever is greater, or $\Delta V_{(BR)}$ = ±2 percent (±5 percent for steady-state operation life) of initial value shall be considered defective. For steady-state operation life, I_R limit (maximum) shall be 2 times the group A limit. Remove defective devices and record the number of failures.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table II herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IIb (JANTXV and JANTX) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb of MIL-S-19500.

Subgroup	Method	Conditions
B3	1026	T_A = +125°C, (see 4.3.2).
B5	3101 or 4081	$R_{\Theta JL}$ ≤ 60°C/W at L = 0.375 inches (9.53 mm), non surface mount devices; $R_{\Theta(JEC)}$ ≤ 20°C/W at L = 0 inches for surface mount devices.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 5 pounds; t = 15 seconds ± 3 seconds, (not applicable to surface mount devices).
C2	2036	Test condition E (not applicable to surface mount devices).
C6	1026	T_A = +125°C, (see 4.3.2). leaded samples from the same lot may be used in lieu of surface mount devices.

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C7	4071	$I_{(BR)}$ = column 3 of table III, $\Delta V_{(BR)}$, T_1 = +25°C ±3°C; T_2 = +125°C ±3°C; maximum limits = column 8 of table III. Sample size: 22 devices, c = 0.
C8		See 4.5.1, 10 pulses, sample size: 22 devices, c = 0.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Maximum peak pulse current (I_{pp}). The peak currents specified in column 7 of table III shall be applied in the reverse direction while simultaneously maintaining a reverse bias voltage of not less than the applicable voltage specified in column 4 of table III. The peak current shall be applied with a current versus time waveform as follows:

a. Pulse current shall reach 100 percent of I_{pp} at $t \leq 8\ \mu s$ and decay to 50 percent at $t \geq 20\ \mu s$ for $t_p = 20\ \mu s$ (see figure 8).

b. Pulse current shall reach 100 percent of I_{pp} at $t \leq 10\ \mu s$ and decay to 50 percent at $t \geq 1$ ms for $t_p = 1$ ms (see figure 9).

NOTE: Tolerance on pulse time shall be ±10 percent.

4.5.2 Clamping voltage V_C. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reverse current	4016	DC method; $V_R = V_{RWM}$ (column 4 of table III)	I_R		Column 5 of table III	µA dc
Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent; $I_{(BR)}$ = column 3 of table III	$V_{(BR)1}$	Column 2 of table III		V dc
Forward voltage	4011	$I_F = 1$ A dc	V_F		1.5	V (pk)
Subgroup 3						
Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent; $I_{(BR)}$ = column 3 of table III, $T_A = -55°C$	$V_{(BR)2}$	Column 9 of table III		V dc
Subgroup 4						
Clamping voltage (see 4.5.2)		$t_p = 1.0$ ms (see 4.5.1b) I_{PP} = column 7 of table III	V_C		Column 6 of table III	V (pk)
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge current	4066	One pulse, half sine wave 8.3 ms; $I_F = 0$, $V_{RWM} = 0$, $T_A = +25°C$, $I_{FSM} = 80$ A (pk)	I_{FSM}			
Electrical measurements		See table II, steps 1 and 2				
Subgroup 7						
Clamping voltage inverse polarity maximum (see 4.5.2)		$t_p = 1.0$ ms (see 4.5.1b) except use forward direction current without prior bias voltage	$-V_C$		Column 11 of table III	V (pk)

1/ For sampling plan, see MIL-S-19500.
2/ Column references are to table III herein.

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TABLE II. Groups A, B, and C electrical end-point measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current	4016	DC method, $V_R = V_{RWM}$ (column 4 of table III)	I_R		Column 5 of table III	µA dc
2.	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table III	$V_{(BR)1}$	Column 2 of table III		V dc
3.	Reverse current	4016	DC method; $V_R = V_{RWM}$ column 4 of table III	ΔI_R 3/		≤ 100 percent of initial reading or 20 percent of column 5 of table III, whichever is greater.	
4.	Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2 percent; I_{BR} = column 3 of table III	$\Delta V_{(BR)1}$ 3/		≤ 5 percent of initial value.	

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see 4.3.2e.

c. Subgroup 6, see table II herein, steps 1, 2, 3, and 4.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 6, see 4.3.2e.

d. Subgroup 8, see table II herein, steps 1 and 2.

3/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE III. Electrical characteristics.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7		Col 8	Col 9	Col 10	Col 11
	$V_{(BR)1}$ at $I_{(BR)}$	$I_{(BR)}$	V_{RWM}	I_R at V_{RWM}	V_C at I_{PP} for t_p = 1 ms	I_{PP} t_p = 20 µs t_r = 8 µs	I_{PP} t_p = 1 ms t_r = 10 µs	$\alpha V_{(BR)}$	$V_{(BR)2}$ minimum at I_{BR} T_A = -55°C	(I_{RM}) maximum dc current T_A = +25°C	$-V_C$ at t_p = 1 ms A(pk) = in accordance with column 7 inverse polarity
	V dc	mA dc	V (pk)	µA dc	V (pk)	A (pk)	A (pk)	%/°C	V dc	mA	V (pk)
1N6461	5.6	25	5	3,000	9.0	315	56	-.03, +.045	5.4	367	-3.5
1N6462	6.5	20	6	2,500	11.0	258	46	+.060	6.2	304	-3.2
1N6463	13.6	5	12	500	22.6	125	22	+.085	13.0	139	-3.8
1N6464	16.4	5	15	500	26.5	107	19	+.085	15.6	63	-3.8
1N6465	27.0	2	24	50	41.4	69	12	+.096	25.1	39	-3.6
1N6466	33.0	1	30.5	3	47.5	63	11	+.098	30.2	34	-3.6
1N6467	43.7	1	40.3	2	63.5	45	8	+.101	40.0	46	-3.5
1N6468	54.0	1	51.6	2	78.5	35	6	+.103	48.5	20	-3.4

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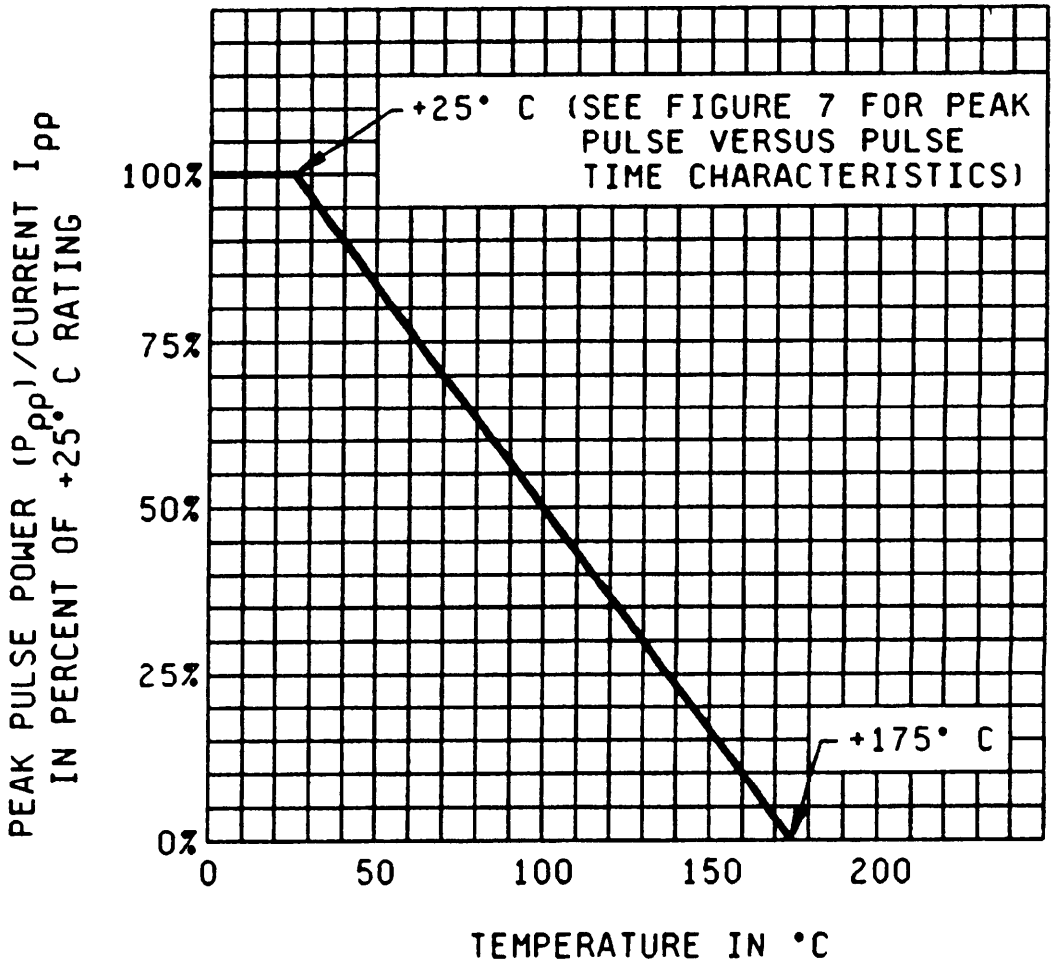


FIGURE 6. Derating curve.

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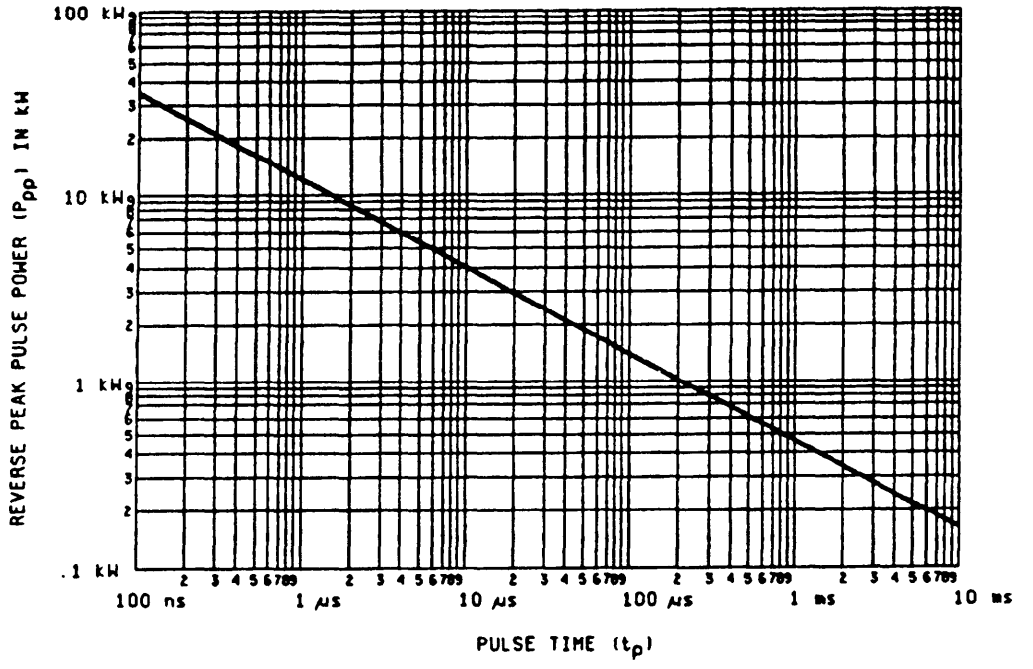


FIGURE 7. Peak pulse power versus pulse time.

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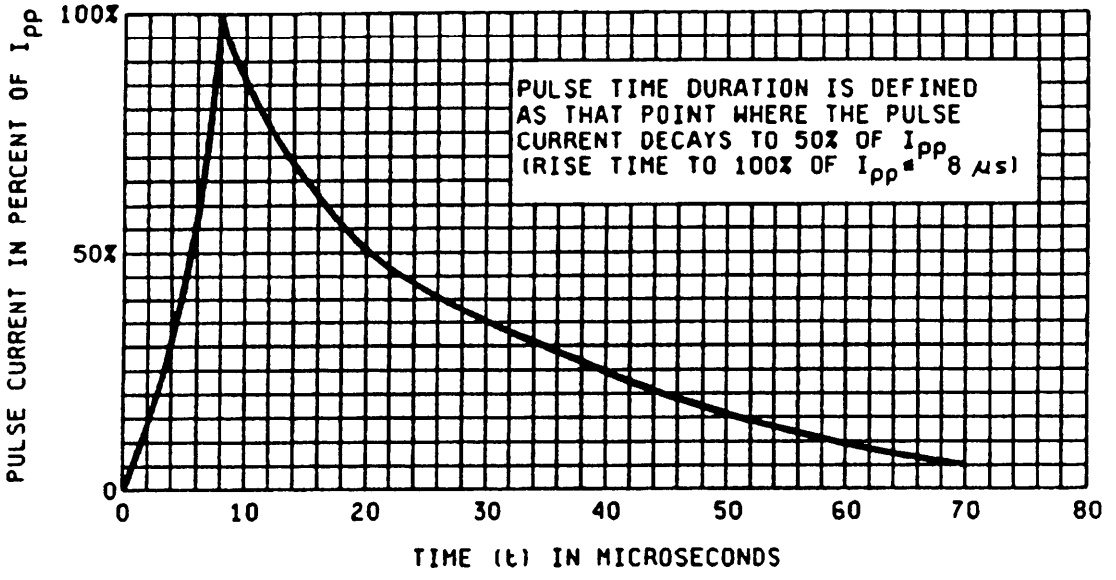


FIGURE 8. Current impulse waveform.

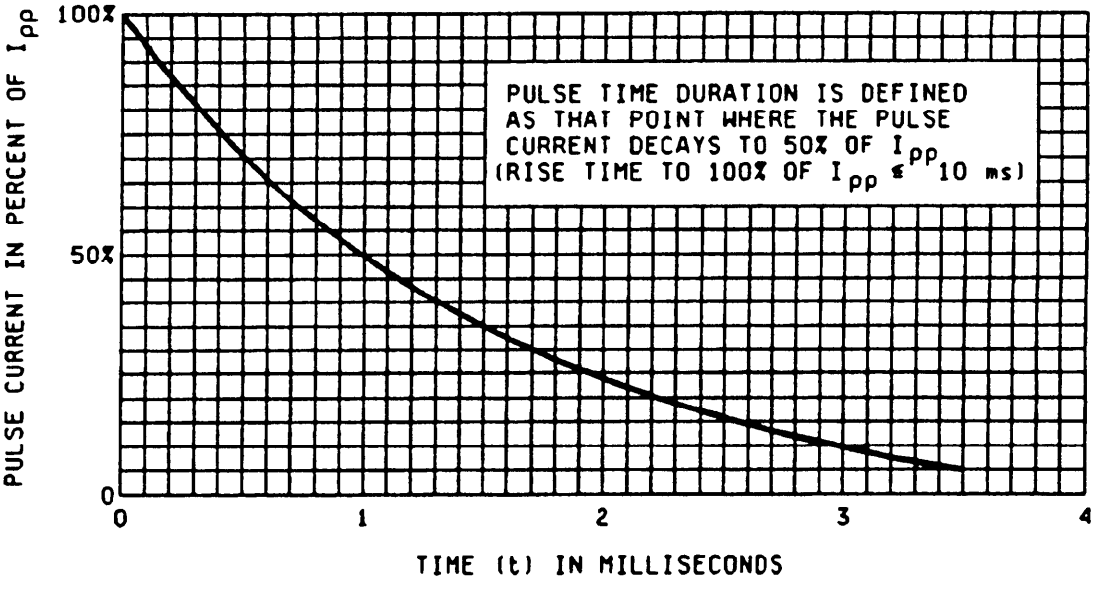


FIGURE 9. Current impulse waveform.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.2).

c. Type designation, product assurance level and for die acquisition, the JANHC and JANKC letter version should be specified.

6.3 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA1N6461) will be identified on the QPL.

JANC ordering information			
	Manufacturer		
PIN	12969	14099	66891
1N6461 through 1N6468	JANHCA1N6461 through JANHCA1N6468	JANHCB1N6461 through JANHCB1N6468	JANHCC1N6461 through JANHCC1N6468

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Air Force - 13, 19, 85
DLA - ES

Preparing activity:
DLA - ES

(Project 5961-1531)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/551B	2. DOCUMENT DATE (YYMMDD) 941229

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, TRANSIENT VOLTAGE SUPPRESSOR TYPES 1N6461 THROUGH 1N6468 AND 1N6461US THROUGH 1N6468US JAN, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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